

Question number	Answer	Additional guidance	Mark
8(a)	A description that includes two of the following points: • taken up by capillary endothelial cells (1) • (virus) replicates in (endothelial cells) causing cell lysis (1) • creating (large) gaps in the capillary wall (1)		(2)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes four of the following points: • antigens are taken up by {macrophages / APC} by phagocytosis (1) • antigens are presented (by macrophages) to T helper cells (1) • activated T helper cells release cytokines to activate B cells (1) • B cells differentiate into plasma cells that release antibodies (1) • B memory cells are developed (1)	ACCEPT antigen engulfed by macrophage	(4)

Question number	Answer	Additional guidance	Mark
8(c)	An answer that includes two of the following points: • sugar molecules are bonded to the vaccine molecules by hydrogen bonds (1) • (therefore when excess) water is added the water molecules form hydrogen bonds with sugar molecules/ vaccine molecules releasing the vaccine (1) • and the vaccine molecules can dissolve in the water (1)	Accept sugar dissolves in water releasing vaccine	(2)

Question number	Answer	Additional guidance	Mark
8(d)	An explanation that includes the following points: • each time the virus replicates mutations may occur (1) • the more mutations that take place the greater the number of changes in the amino acid sequence of the virus coat proteins (1) • immune response is specific and will not produce antibodies against the Ebola virus (1)	Accept during replication of the virus mutations occur	(3)

Question number	Answer	Additional guidance	Mark
8(e)(i)	A description that includes the following points: • Ebola virus antigen is a protein (1) • isolate / synthesise the gene coding for Ebola virus antigen (1) • insert the gene into a vector (1) • description of method of putting the vector into the insect cell (1)	Plasmid/bacterial or viral vector, liposome Micro-injection / gene gun / heat shock	(4)

Question number	Answer	Additional guidance	Mark
8(e)(ii)	A description that includes the following points: • it is safer to use genetically modified insect cells than the Ebola virus (1) • insect cells are eukaryotic cells / contain endoplasmic reticulum and Golgi apparatus (1) • when the vaccine protein is made in insect cells it will be (properly) {glycosylated / post transcriptionally modified} (1) • (insect cells) have a low mutation rate (1)	Accept less chance of a mutation	(3)

Question number	Answer	Additional guidance	Mark
8(f)	An explanation that includes the following points: - different regions of nucleic acid code for the stem and the head (1) - there is only one version (of nucleic acid) coding for the stem but many versions for the head (1) May need to accept explanations in terms of post-transcriptional modification - splicing of the gene for haemagglutinin (1) - same exons used for the stem and different exons used to code for head (1)	ALLOW There are more mutations in the gene(s) for the head than in the gene(s) for the stem (for 2 marks)	(2)

